Ben Bubnick

DATA ENGINEER · DATA SCIENTIST · ANALYTICS SPECIALIST

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Summary

Hands-on engineer specializing in Python on UNIX-like systems (Linux/macOS), production ML/MLOps, and large-scale data pipelines. Heavy daily use of Terminal/Bash/Zsh, SSH, and POSIX tooling to build and operate services. Built forecasting and optimization solutions (e.g., surgery duration prediction, hospitalization risk) and led end-to-end data platforms in AWS and Azure. Strong at turning ambiguous operational problems into real-time services with CI/CD, observability, and documentation. Comfortable collaborating across engineering, data science, and ops—ready to lead AI application development and model integration in high-performance, resource- and cost-sensitive environments.

Experience

GrowData Analytics

AWS: S3, Glue, Athena, Lambda, CloudWatch, dbt, Snowflake, Ruby, RubyLLM, Tesseract OCR, GPT 4o mini, Cladue Opus/Sonnet, Rubocop, RSpec

PRINCIPAL DATA ENGINEER *Oct. 2025 - Present*

Led data engineering work spanning source analysis, data modeling, ETL development, identity resolution, and QA to deliver standardized healthcare datasets for reporting and downstream integration.

- Led end-to-end design and delivery of healthcare data integration pipelines, converting operational source data into curated analytics-ready models for reporting and downstream use.
- Conducted feasibility analysis of source-system structures and implementation constraints to shape integration scope, modeling strategy, and delivery approach.
- Designed and refined core domain models for organization, office, caregiver, patient, and crosswalk entities to support standardized healthcare data workflows.
- Structured target models using FHIR-like concepts where appropriate, improving consistency and extensibility for downstream integration and analytics use cases.
- Developed reusable ETL and transformation logic across development and sandbox environments using SQL and warehouse-native modeling practices.
- Implemented source-to-target standardization patterns for null handling, lifecycle fields, sentinel dates, and source-specific variation.
- Defined surrogate key, natural key, and crosswalk strategies to enable stable identity resolution across related source systems.
- Designed and executed comprehensive QA and data-integrity test packs covering row counts, duplicates, nulls, referential integrity, reconciliation, and exception handling.
- Resolved data-quality and mapping issues through targeted test-case analysis and review of failing source and target records.
- Documented mapping logic, model assumptions, QA expectations, and implementation standards to improve delivery reliability and reuse.
- Collaborated with client and internal stakeholders to resolve source ambiguity, refine requirements, and translate business needs into production-ready data assets.
- Supported future scalability by establishing reusable patterns for onboarding additional sources and extending integration workflows.

PRINCIPAL ML ENGINEER *Sep. 2024 - Oct. 2025*

Built and shipped an OCR-to-analytics product for small farms; owned model integration, CI/CD, testing, and auditability on UNIX-like systems.

- Productized OCR → structured data pipeline for farm invoices (handwritten/printed) using Tesseract + rule-based post-processing; shipped as a small Python/Ruby service with containerized deployments and versioned configs.
- Built validation & reconciliation layers (schema checks, cross-field rules, idempotent re-runs) and produced audit logs to support bookkeeping/compliance workflows.
- Implemented CI/CD (unit tests, linting, fixtures, golden-file tests) and release automation; documented runbooks and failure modes (timeouts, low-confidence OCR, retries).
- Delivered a lightweight analytics layer (usage metrics, error taxonomy, latency/throughput dashboards) to guide feature tuning and cost control.
- Owned infra as code for a minimal AWS footprint (S3 for artifacts, Lambda/Cron for jobs, CloudWatch alerts); operated via UNIX-like CLI workflows (Bash/Zsh, Makefile targets).

Health Data Movers

AWS: S3, Snowflake, SQLMesh, Python, SQL, Docker

DATA CONSULTANT *Mar. 2025 - Jul. 2025*

Developed a reusable ETL pipeline to streamline data onboarding for a healthcare revenue cycle platform.

- Built a reusable ingestion & transformation framework in SQLMesh/Snowflake (templated models, data contracts, CI checks, promotion gates), enabling consistent, repeatable onboardings.
- Added observability & auditability (row-count/parity checks, schema drift alerts, reconciliation logs) and wrote runbooks for deterministic re-runs and incident handling on UNIX-like systems.
- Shipped a lightweight forecasting/alerting pattern (Python service + Docker + Actions) for operations signals; packaged with tests and deployment automation.
- Created testable, modular SQLMesh models, reducing regression errors and accelerating iteration cycles.
- Standardized and documented transformation logic across clients in order to cut onboarding documentation.
- Established validation and audit checkpoints (e.g., row counts, null checks, referential integrity).
- Translated business requirements into technical specifications to improve delivery predictability.
- Built detailed integration playbooks to be used as the internal standard for onboarding new vendors and partners.

Arkos Health

AWS: S3, Athena, Redshift, Glue, dbt, PySpark, Docker + AWS CLI, Tableau, Jupyter

SENIOR DATA ANALYST *Jun. 2023 - Jan. 2025*

Built a scalable AWS data lake with automated CI/CD, metadata reporting, and integrated analytics tools.

- Stood up an AWS data lake with CI/CD (dbt + Git flows), cost-aware partitioning, and job scheduling; reduced deployment time 50% and improved query performance with targeted model refactors.
- Implemented monitoring/alerting & on-call runbooks (metrics, logs, failure taxonomies) to lower MTTR and improve data reliability.
- Mentored engineers on code reviews, testing, and pipeline design; authored internal standards that lifted team delivery quality.
- Integrated Jupyter into reporting, boosting report interactivity and increasing stakeholder satisfaction by 30%.
- Created comprehensive data lake documentation, reducing onboarding time for new team members by 40%.
- Implemented Jira and GitHub, enhancing task tracking and team coordination by 25% within three months.
- Enhanced cross-departmental collaboration, increasing data-driven decision-making by 15% in six months.
- Mentored junior team members, improving their performance metrics by 30% in six months.
- Led goal-setting sessions, improving goal alignment and tracking accuracy by 30% in the first quarter.
- Initiated stakeholder feedback sessions, improving deliverable relevance and accuracy by 25% in three months.

Merative
Azure: Data Factory, Blob Storage, Functions, SQL DB, Synapse Analytics, Python, DevOps

SENIOR DATA SCIENTIST *Jul. 2022 - Mar. 2023*

Launched a new Azure platform to replace HDFS systems to modernize data infrastructure.

- Led HIPAA-compliant deidentification initiative across 200+ million patient records.
- Designed and implemented an automated testing framework for ETL pipelines for validation coverage.
- Validated and optimized Azure-based data workflows, ensuring continuity and fidelity during the transition from legacy HDFS systems to cloud architecture.
- Conducted end-to-end system testing for Azure Data Factory pipelines, Blob Storage configurations, and downstream Synapse integrations.
- Collaborated with engineering teams across disciplines to refine architectural designs.
- Documented test plans and platform migration procedures for improving onboarding for future QA engineers.
- Built regression test suites and data comparison tools, detecting schema drift and transformation errors early in the development lifecycle.
- Ensured compliance with data governance and security protocols to reduce audit risks and support HITRUST readiness.

IBM
Hadoop/HDFS/CDH: Spark, Hive & Impala, Kafka, Python, JRuby, SQL, Pig, Docker + Gradle

SENIOR DATA SCIENTIST *Nov. 2021 - Jul. 2022*

Built AI tools while leading analytics delivery and large-scale data validation efforts.

- Created AI Surgical Scheduling Optimization tool currently evolving into new offering.
- Led delivery on tailored analytics and reporting project for client administrative data.
- Developed data migration validation analysis on 150+ billion patient records.
- Developed surgical smart case card using combined CNN/RNN model.

SENIOR DATA CURATION ARCHITECT *Dec. 2020 - Nov. 2021*

Led large-scale clinical data and NLP initiatives and delivering ML solutions.

- Developed clinical data integration of 100+ billion structured & semi-structured records.
- Developed topic modelling project on nearly 6 million semi-structured provider notes data.
- Curated 25 pages of workflow documentation and mentored new hires in AI tuning process.
- Created COVID-19 hospitalization prediction tool with Machine Learning algorithms.

DATA CURATION ARCHITECT *Nov. 2017 - Dec. 2020*

Led enterprise data integration initiatives and mentoring programs.

- Managed multi-org enterprise integration with over 70 interconnected data sources.
- Developed new data validation and review process that reduced total rework by 25%.
- Reduced mapping rework by 45% with through new knowledge-driven initiatives.
- Led mentoring program to increase 4,500 billable hours over two year period.

SOFTWARE DEVELOPER *Sep. 2016 - Nov. 2017*

Developed internal resources that reduced costs and improved onboarding efficiency.

- Reduced time spent on knowledge transfer by 80% by rewriting project documentation.
- Eliminated need for a $30,000 annual contract by developing data curation tool.
- Reduced cost-per-hire by $5,210 by re-developing education program.

DATA SCIENTIST *Nov. 2015 - Sep. 2016*

Led integration efforts and process automation initiatives.

- Managed team of 12 remote data science contractors for 36 integration projects.
- Created over 125 new knowledge base articles, making up 30% of the team's online content.
- Developed automated data integration tool to cut out 28% unnecessary development time.

Explorys
Hadoop/HDFS: Ruby, Pig, React, NodeJS, JavaScript, SQL, Oracle, Jenkins

DATA SCIENTIST *Jun. 2015 - Nov. 2015*

Built ETL pipelines on distributed data systems to support clinical data integration and population analytics.

- Principal integration specialist for 10+ client database warehouses.
- Performed troubleshooting on production issues across multiple environments.
- Developed cron-based widget to define blackout periods for client data pulls.

Multiple Institutions

ADJUNCT PROFESSOR *Aug. 2010 - May. 2019*

Taught physics & astronomy lecture series and lab courses.

- Consistently exceeded student evaluation benchmarks, surpassing institutional feedback goals by an average of 16% in 2017 and 17% in 2018, reflecting strong teaching effectiveness and student satisfaction.
- Improved student comprehension and course navigation by reorganizing online instructional materials, resulting in a 28% reduction in reported confusion and fewer clarification requests during office hours.
- Developed and published over 700 hours of recorded lectures and lab content, enhancing accessibility and supporting flexible, on-demand learning for both traditional and non-traditional students.
- Implemented Just-in-Time Teaching (JiTT) strategies for underperforming students, leading to an average grade improvement of 18% by the end of term.
- Designed interactive lab lectures based on the Interactive Lecture Demonstration (ILD) method, increasing student engagement and concept retention in physics instruction.
- Promoted collaborative learning through Peer Instruction and ILD techniques, effectively addressing real-time student misconceptions during physics lab sessions.

AmTrust Financial Services
VB.NET, ASP.NET, C#, CSS, Javascript, HTML5, SQL, PL/SQL

SOFTWARE ENGINEER *Mar. 2014 - Jun. 2015*

Developed MVC components and web forms for insurance policy software.

- Led development on core insurance policy software, serving as principal developer for a new product offering and contributing to two major architectural initiatives in 2015.
- Analyzed and optimized performance across 65+ ISO rating-based web forms using R, identifying bottlenecks and reducing page load times.
- Championed and implemented a unit testing framework for multi-tier MVC components, improving code quality and reducing deployment errors across the dev team.

PPG Industries
VB.NET, C#, SQL, C++, FORTRAN

COLOR SCIENTIST *Apr. 2012 - Jun. 2013*

Developed MVC components and web forms for insurance policy software.

- Improved color matching classification algorithm accuracy by 40% by developing and applying advanced numerical methods, enhancing product performance across automotive coatings.
- Developed cross-instrument correlation software to support a risk analysis initiative, enabling consistent color measurement across varying hardware platforms and improving QA reliability.
- Supported the creation of an industry-leading automated color matching facility, developing numerical tools and methods to improve formulation quality and pigment data analysis in high-variance OEM color environments.

Professional Recognition

HDM, SPECIAL RECOGNITION (*2025*), Recognized for exceptional performance and commitment in client delivery.

Arkos, ARKOS HEALTH BI STAR (*2023*), Recognition for Achieving Expeditioner Rank, Salesforce Trailhead.

Merative, MANAGER RECOGNITION FOR TRANSITION LEADERSHIP (*2022*), Recognized for systems migration efforts.

IBM, GOLD CHAMPION (*2018 - 2022*), Recognized for 1,477 classroom hours delivered and 43 certificates earned.

IBM, LAB SERVICES AWARD (*2018*), Recognized for coordinating internal teams to deliver solutions on time.

IBM, EMINENCE AND EXCELLENCE AWARD (*2017*), Recognized for developing a new data curation tool.

PPG, COLOR LAB AUTOMATION AWARD (*2012*), Recognized for expanding data collection and analysis reporting.

Volunteer

HSEA
ETL Brigade *Feb. 2025 - May. 2025*
Built a reusable PostgreSQL data model to help community health organizations prototype analytics tools for claims, visits, and eligibility.

CSforAll
Instructor *Oct. 2018 - May. 2019*
Our goal was to provide computer science instruction in all of the District's high schools, with a special emphasis on students who do not have access to a computer or a stable Internet connection at home.

HIT in the CLE
Coach *Mar. 2017 - Mar. 2019*
We were using a data science competition to educate students understand the potential of a career in health information technology and to assist IT companies in Northeast Ohio in developing a talent pipeline.

Code for America
ETL Guild *Jan. 2015 - Jun. 2015*
The initiative was brought about to apply data science tools to improve and expand public record access.

T.C.P. World Academy
Tutor *Sep. 2008 - May. 2009*
Our goal was to teach students how to understand the value of their learning experience and becoming academically involved by participating in society simulation activities for higher learning.

Selected Projects

farm-invoice-parser
2025
Ruby, RubyLLM, Tesseract OCR, OpenAI/Anthropic model APIs

Built a RubyLLM-based tool to convert handwritten farm invoices into structured transaction records for small farms.

healthcare-dbt-demo
2025
dbt, Data Vault 2.0, DuckDB/Postgres, SQL modeling

Created a minimal, runnable Data Vault 2.0 demo for healthcare claims data using dbt.

HL7-ingestion
2023
Java, Docker, HL7, FHIR

Developed a healthcare interoperability project to convert HL7 messages into longitudinal patient records.

tuva-postgres
2025
Python, Shell, PostgreSQL, Makefile

Built utilities to convert Tuva seed data into a PostgreSQL-ready format.

quality_measures_engine
2023
Python, healthcare quality logic, analytics automation

Built a Python-based project focused on healthcare quality-measure processing and evaluation workflows.

DETL-design-challenge
2023
health data ETL, transformation design, data validation, workflow automation

Designed and implemented a health data transformation and loading approach to automate part of the ETL process.

Talks

AI Hackathon 2020: Surgery Scheduling Optimization *Dec. 2020*
"An algorithm uses past utilization and current waitlist info to predict surgery duration. IBM

REACH Initiative: IBM Watson Care Insights *2019*
Analyze patient records and real-time data via HL7 feeds, integrate insights into physician's EHR workflow. IBM

Content & Data Analytics Learning Exchange *2018*
Introduction of implementation of machine learning algorithms using real-world health data analytics. IBM

2018 Innovation Research Interchange Fall Networks Conference *Sep. 2018*
Using NLP to extract insight from unstructured data and transform it from a data lake. IRI

Cross Team Training Series *2016*
A 7-part series on using Hadoop for data ingestion that shows advantages of data lakes. Explorys

Publications

Reproducible Postgres load of Tuva seed datasets *Aug. 2025*
Reproducible Postgres load of Tuva seed datasets with robust validation. HSEA

Surgical Scheduling Optimization and Procedure Duration Prediction [a] *Dec. 2020*
Improvement on traditional fixed ratio methods for total procedure time estimation. IBM

Process and Requirements for CCD Data Integration *Oct. 2017*
Processes and requirements for overcoming the many difficulties with CCD data curation. IBM

AI Paint Formula Prediction from Spectroscopic Measurements [b] *May. 2015*
Naive bayesian color and effects matching algorithm of complex automotive paint formula. shinyapps.io

Characterization of Surface Coatings using RT Models for Color Matching *Feb. 2013*
Approximation of Chandrasekhar RT model for atmospheric scattering applied to color matching. USPTO

Massive Stellar Clusters in the Disk of the Milky Way Galaxy [c] *Dec. 2010*
Spectroscopic and Photometric analysis of two open clusters in the galactic plane. OhioLINK ETD

VizieR Online Data Catalog: JHK photometry in Cluster [BDS2003] 107 *May. 2008*
Spectra were obtained 2005 June 1415 at the 3.0 m NASA Infrared Telescope Facility (IRTF) on Mauna Kea, Hawaii, using SpeX. VizieR

Near-Infrared Spectroscopy of Candidates Members of the Galactic Cluster [BDS2003] 107 *Feb. 2008*
New near-infrared classification spectra for nine candidate members of the galactic cluster [BDS2003] 107 (Cluster 107). PASP

A Novel Method for Low-Toxic Photo-Etch Printing [d] *May. 2003*
Application of aquatint etching techniques to photo-etch procedures. MAPC

Education

2022	**PGCert.**, Data Science Professional[a]	*IBM*
2015	**PGCert.**, Data Science[b]	*Johns Hopkins University*
2010	**Master of Science**, Astrophysics[c]	*University of Cincinnati*
2004	**Bachelor of Fine Arts**, Printmaking[d]	*Ohio University*